6.1 Exponential Growth and Decay Functions

Name _____ Period ____ Date _____

Algebra II 6.1 (30-minute plan) · Student sheet

I can…

- identify and evaluate exponential functions.
- tell exponential growth from decay and find the percent rate of change.
- write and use exponential models, including compound interest.

Vocabulary · fill in the missing word

Exponential function — $y =$ _____, where $a \neq 0$, $b > 0$, and $b \neq 1$

Exponential growth / decay — _____

Asymptote — a line a graph gets _____ and closer to; for $y = ab^x$ it is $y = 0$

Growth / decay model — $y = a(1 + r)^t$ / $y = a(1 - r)^t$, where r is the _____ rate as a decimal

Compound interest — $A =$ _____

Part 1 · Notes and examples (fill in each note, then work the example)

Exponential function: An **exponential function** has the form $y =$ _____, where $a \neq 0$, $b > 0$, and $b \neq 1$.

Example 1

Evaluate $y = 4(3)^x$ for $x = -2$ and $x = 2$.

Growth or decay: If $b > 1$, the function shows exponential _____. If $0 < b < 1$, it shows exponential _____.

Asymptote: The graph of $y = ab^x$ has a horizontal asymptote at _____.

Example 2

Determine whether each function represents exponential growth or exponential decay.

a. $y = 2(0.6)^x$ b. $y = 0.5(1.25)^x$

Growth and decay models: Growth: $y = a(1 + r)^t$. Decay: $y =$ _____.

Compound interest: $A =$ _____

Example 3

The amount y (in milligrams) of a medicine in your bloodstream t hours after a dose is $y = 200(0.85)^t$. Is it growth or decay? Find the percent rate of change. How much is left after 3 hours?

Part 2 · Partner practice

Partner practice

The number of users of a website t years after it launched is $y = 1500(1.03)^t$. Is it growth or decay? Find the percent rate of change.